can make the calculations, quantum mechanics allows us to calculate only the *probability* of an event. This absence of determinism at a small scale is intrinsic to nature. An electron is not obliged by nature to move towards the right or the left; it does so by chance. The apparent determinism of the macroscopic world is due only to the fact that the microscopic randomness cancels out on average, leaving only fluctuations too minute for us to perceive in everyday life.

Dirac's quantum mechanics thus allows us to do two things. The first is to calculate *which* values a physical variable may assume. This is called 'calculation of the spectrum of a variable'; it captures the *granular* nature of things. When an object (atom, electromagnetic field, molecule, pendulum, stone, star, and so on) interacts with something else, the values computed are those which its variables can assume in the interaction (*relationism*). The second thing that Dirac's quantum mechanics allows us to do is to compute the *probability* that this or that value of a variable appears at the next interaction. This is called 'calculation of an *amplitude of transition*'. Probability expresses the third feature of the theory: *indeterminacy* – the fact that it does not give unique predictions, only probabilistic ones.

This is Dirac's quantum mechanics: a recipe for calculating the spectra of the variables and a recipe for calculating the probability that one or another value in the spectrum will appear during an interaction. That's it. What happens between one interaction and the next is not mentioned in the theory. It does not exist.

The probability of finding an electron or any other particle at one point or another can be imagined as a diffuse cloud, denser where the probability of seeing the particle is stronger. Sometimes it is useful to visualize this cloud as if it were a real thing. For instance, the cloud that represents an electron around its nucleus indicates where it is more likely that the electron appears if we look at it. Perhaps you encountered them at school: these are the atomic 'orbitals'.[fn22]

The efficacy of the theory soon proves extraordinary. If today we build computers, have advanced molecular chemistry and biology, lasers and semiconductors, it is thanks to quantum mechanics. For a certain number of decades it was as if it were Christmas every day for physicists: for every new problem, there was an answer which followed from the equations of quantum mechanics, and it was always the correct answer. One example of this will suffice.

The matter surrounding us is made up of a thousand different substances. During the nineteenth and twentieth centuries chemists understood that all these different substances are just combinations of a relatively small number (less than a hundred) of simple elements: hydrogen, helium, oxygen, and so on – to uranium. Mendeleev put these elements in order (according to weight)